

Question_5:

Code:

```
1  /*
2  QUESTION_5:
3  Delete alternate nodes from a singly linked list.
4  The deletion starts from the SECOND node.
5  */
6  #include <iostream>
7  using namespace std;
8
9  struct node {
10     int data;
11     node *next;
12 };
13 void insert_Data(node *&head, int x) {
14     node *p = new node{x, NULL};
15
16     if (!head)
17         head = p;
18     else {
19         node *t = head;
20         while (t->next)
21             t = t->next;
22         t->next = p;
23     }
24 }
25 void delete_Alt(node *head) {
26     while (head && head->next) {
27         head->next = head->next->next;
28         head = head->next;
29     }
30 }
31 int main() {
```

```
32     int n;  
33     cin >> n;  
34  
35     node *head = NULL;  
36  
37     for (int i = 1; i <= n; i++)  
38         insert_Data(head, i);  
39  
40     delete_Alt(head);  
41     while (head) {  
42         cout << head->data << " ";  
43         head = head->next;  
44     }  
45     return 0;  
46 }
```

Output:

Test Case_1:

```
10  
1 3 5 7 9  
PS C:\Users\Prabin Yadav
```

Test Case_2:

```
20  
1 3 5 7 9 11 13 15 17 19  
PS C:\Users\Prabin Yadav\Documents\
```